

Quantum Random Walks for Market Microstructure

Phase 6 Technical Report

BTCUSDT event data | June 12, 2026
Reproducible exploratory benchmark

Engineering conclusion: pipeline complete; no validated QRW superiority.

Abstract

This report evaluates a causally calibrated discrete-time quantum random walk for short-horizon market microstructure. It compares QRW Adaptive with three classical random walks, GARCH(1,1), and GBM under one chronological benchmark.

The active sample has 1,869,214 training rows and 1,246,144 later holdout rows. CRW Correlated ranks first, while QRW ranks 7.

The software and artifact checkpoint passes, but the single-window evidence does not establish predictive or distributional superiority.

1. Motivation and Research Question

Market data arrive as discrete, dependent events with changing liquidity and heavy tails. A QRW offers interference and a controlled coherent-to-diffusive transition that classical random walks do not possess.

The empirical question is whether those extra mechanisms improve fit under a fair chronological comparison. The project does not claim that market mechanics are quantum.

2. QRW Formalism

A coin operation acts on a two-state direction space, followed by a conditional shift on a one-dimensional position lattice. Position probabilities are obtained by marginalizing squared amplitudes over the coin state.

The coherent Hadamard walk spreads ballistically. Basis dephasing damps off-diagonal density-matrix entries while preserving trace and populations, producing a transition toward classical diffusion.

3. Market Mapping

Order-book imbalance, tick direction, imbalance change, and absolute imbalance form the adaptive directional signal. Trade intensity modulates event-level dephasing.

The adaptive operator remains unitary. Its directional signal is encoded in phase, which is why magnitude-squared coin entries remain balanced.

4. Data and Causal Controls

Active artifact: data\features\features_BTCUSDT_2026-06-12.parquet. The final study uses only the causal-cleaned June 12, 2026 window.

Historical June 1-7 derived artifacts were invalidated after detection of a one-tick look-ahead in the outlier filter. They may not support final conclusions until rebuilt.

Calibration is chronological. Structural validation is not included in refitting, and the later bias update does not reuse the structural warmup.

5. Benchmark and Statistical Protocol

The benchmark uses 3,000 paths per model, 500 simulation steps, and random seed 2026.

Evaluation covers distribution tests, variance scaling with bootstrap intervals, return autocorrelation, tail metrics, and an eight-metric rank scorecard.

P-values are corrected within families using the Benjamini-Hochberg procedure. AIC and BIC are interpreted only within matching likelihood families.

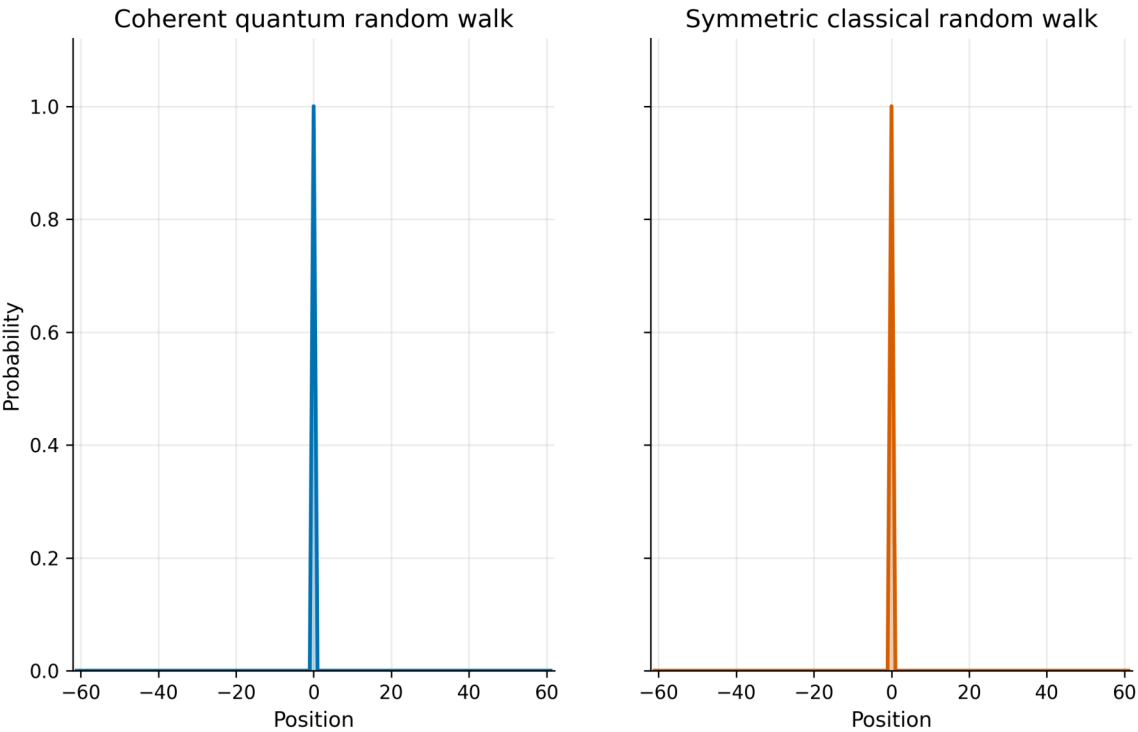
6. Model Implementations

QRW Adaptive uses a unitary phase-adaptive coin and intensity-dependent basis dephasing. Classical baselines include simple, biased, and correlated random walks.

GARCH(1,1) supplies conditional volatility clustering and GBM supplies a continuous lognormal diffusion baseline. All models use the same training cutoff and test horizon.

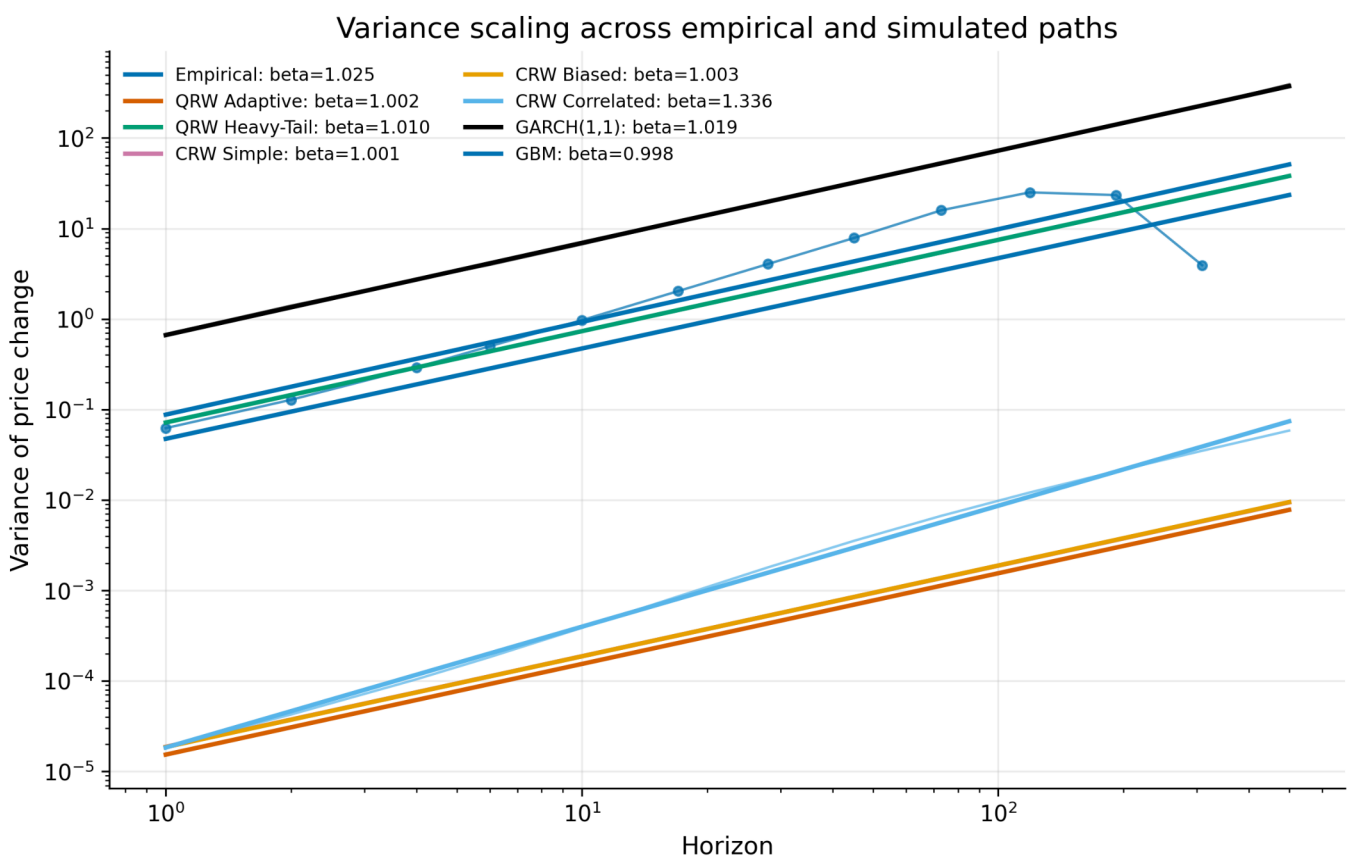
Probability Evolution

Probability evolution at $t = 0$



Coherent QRW interference and ballistic spread contrast with the diffusive classical binomial distribution.

Variance Scaling



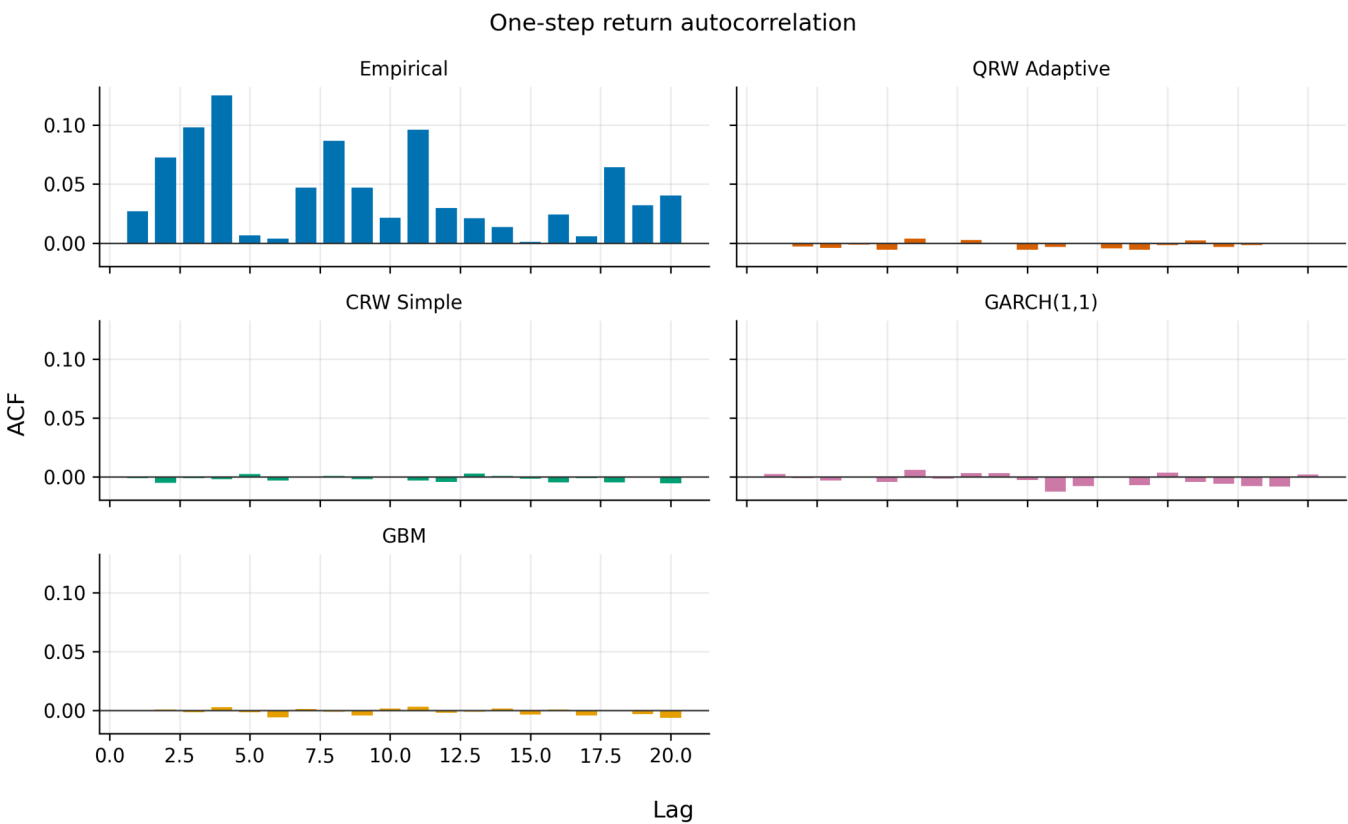
Regression slopes summarize how price-change variance grows with horizon for the empirical path and every simulated model.

Return Distribution Comparison



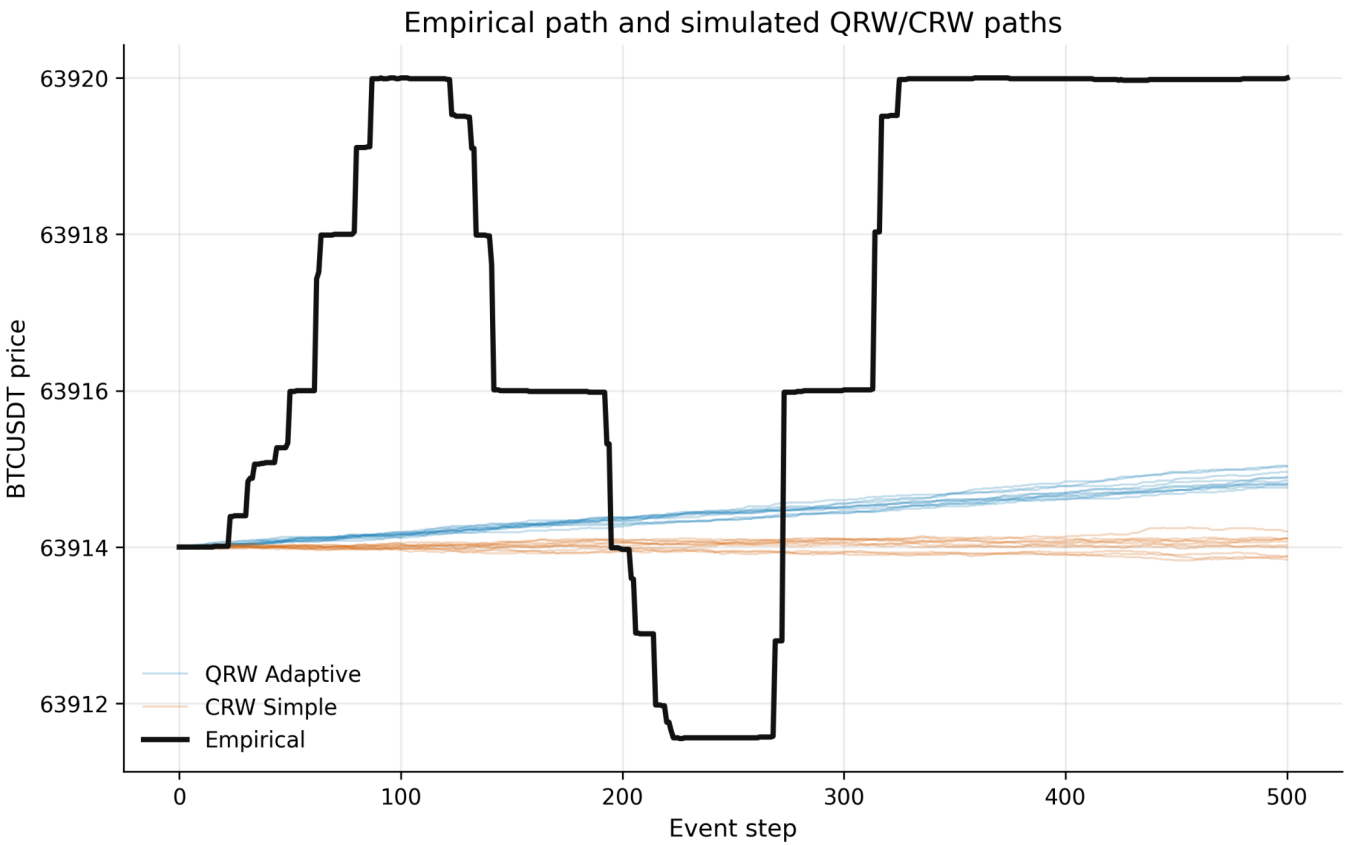
Returns are standardized per series, so the comparison concerns distributional shape rather than volatility scale.

Autocorrelation Comparison



The empirical series has dependence that no single simulated baseline reproduces completely.

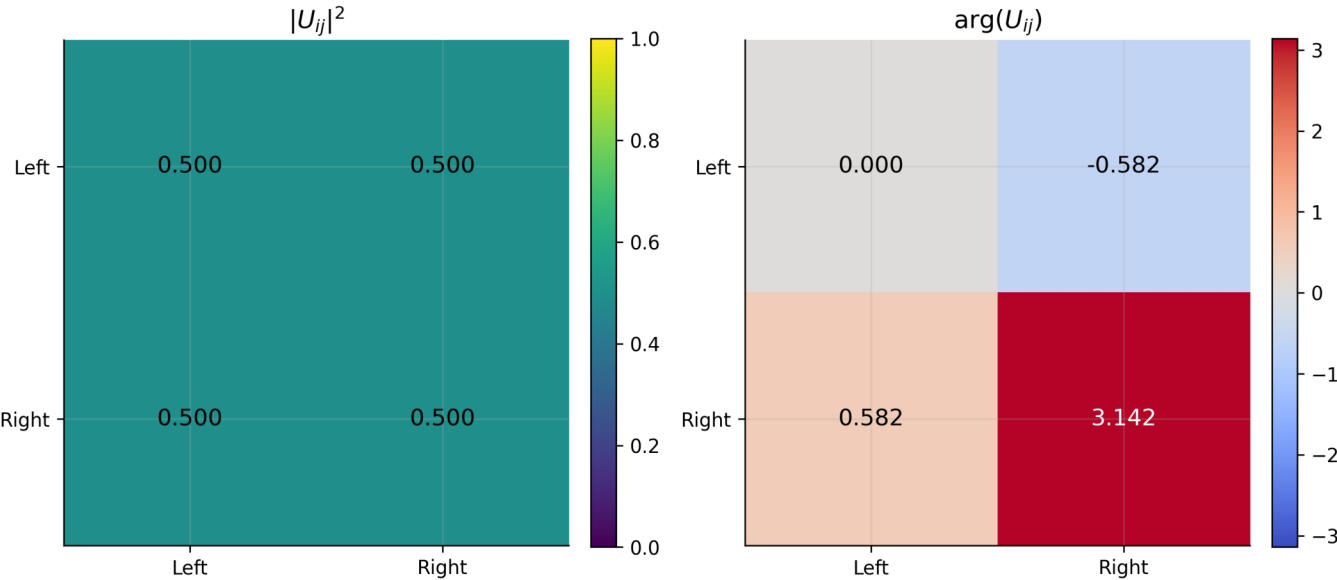
Sample Paths



Direct path overlays reveal scale and directional differences that are compressed by aggregate scores.

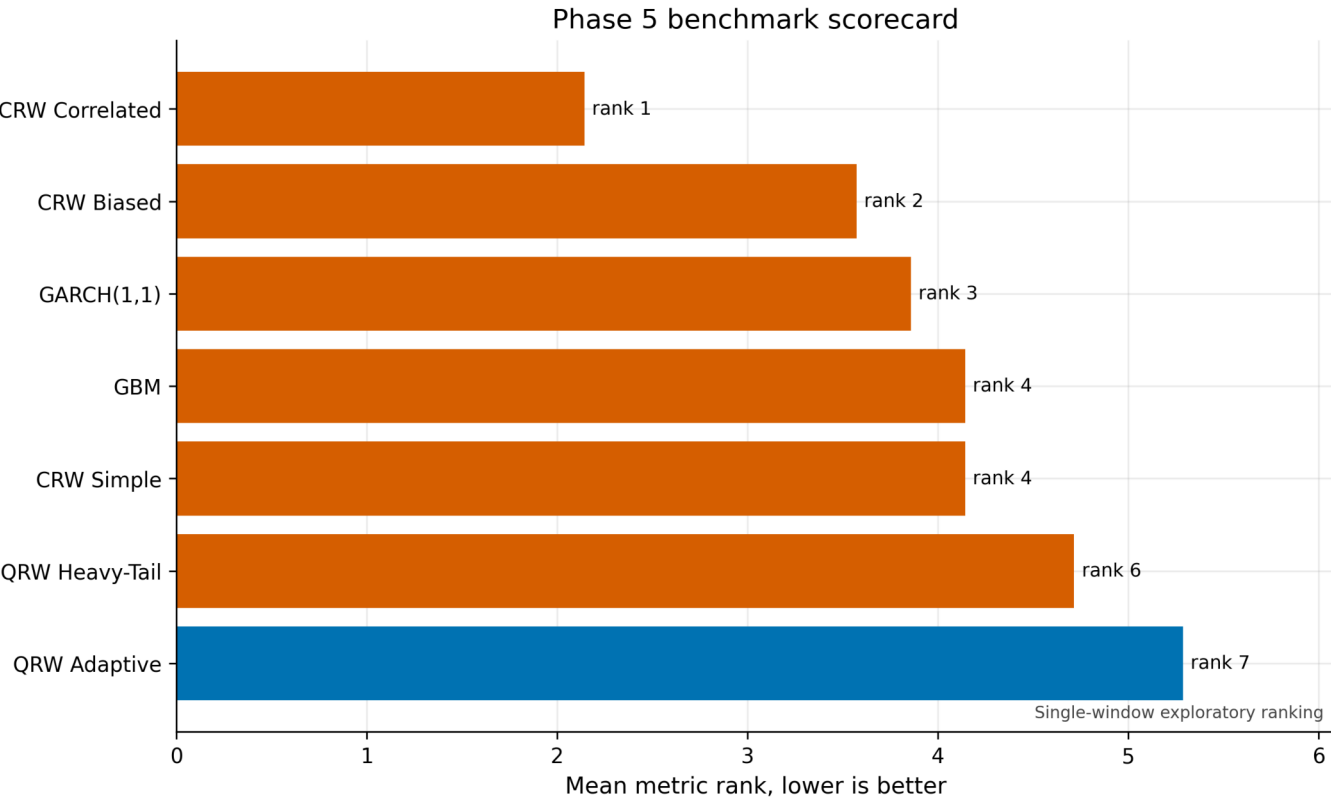
Adaptive Coin Operator

OBI-adaptive coin at OBI=0.75, alpha=0.82
Directional information is phase-encoded; magnitudes remain unitary.



The operator keeps balanced magnitudes while market information changes complex phases.

Benchmark Scorecard



CRW Correlated is first overall. QRW is rank 7; this is an exploratory rank average.

Quantitative Result Table

Quantity	Observed value
Top model	CRW Correlated
Top mean rank	2.143
QRW overall rank	7
QRW mean rank	5.286
QRW KS p-value	5.459e-04
QRW KS p-value, BH	9.553e-04
Empirical variance beta	1.2847
QRW variance beta	1.0026
QRW beta 95% CI	[0.9841, 1.0183]
Empirical tail index	2.4748
QRW tail index	1.598e+06
QRW ACF MSE	0.003275

All values are exploratory and come from one June 12, 2026 window.

7. Discussion

CRW Correlated has the best aggregate scorecard rank. QRW ranks 7, performing well on variance scaling and ACF distance but poorly on the one-step distribution test.

Zero-inflated fixed-tick local moves yield tails that are much thinner than the empirical sample. GARCH captures heavy tails more plausibly but loses rank on other metrics.

Phase 3 is mixed: QRW wins the final holdout but loses the pooled walk-forward comparison against a fair affine baseline.

8. Limitations

- The study uses one short event window and limited real synchronized order-book coverage.
- The adaptive calibration is data constrained, and the rank scorecard compresses dependent metrics to an equal weight average.
- No current artifact is a fresh untouched confirmatory holdout. The dashboard and visualizations are exploratory interfaces, not additional evidence.

9. Conclusion and Future Work

Phase 6 completes the reproducible visualization and reporting layer. The defensible conclusion is an engineering pass with no validated QRW superiority.

Next work should freeze the protocol before observing new labels, collect at least 20 fresh UTC days with synchronized LOB data, and run one confirmatory analysis.

Further model work may study regularized learned coins, two-dimensional multi-asset QRWs, and continuous-time walks against equally flexible classical point-process baselines.

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